WILLIAM SENTOSA

+1(217) 721-7885 ◇ San Jose, CA, USA ◇ ◇ williamsentosa.sentosa@gmail.com ◇ sentosa2@illinois.edu
https://williamsentosa.github.io

EDUCATION

University of Illinois Urbana-Champaign, USA *Aug 2018 - May 2026*
Ph.D. in Computer Science, area: Systems and Networking
Advisor: P. Brighten Godfrey

Bandung Institute of Technology, Indonesia *Aug 2013 - Oct 2017*
B.S. in Computer Science GPA: 3.83/4.00

PROFILE

I am a networking systems researcher and engineer. My previous work spans a broad range of networking systems, including **low-latency networking, transport protocols, multipath networking, performance modeling, wide-area networks, and network measurement**. My work centers on three main topics: (1) addressing fundamental low-latency networking challenges, including application latency in web browsing and video streaming, as well as ISP and WiFi latency; (2) designing and implementing multipath transport protocols; and (3) understanding and modeling cellular networks behavior for mobile applications.

THESIS

Low-Latency Network Transport Services over Heterogeneous Paths
William Sentosa
My dissertation tackles a fundamental networking challenge: achieving both high bandwidth and low latency in wireless networks. Next-generation networks such as 5G and 6G offer heterogeneous paths optimized for either high bandwidth or low latency to address the inherent trade-off between the two on a single path. My dissertation aims to overcome this trade-off by enabling applications to leverage multiple paths simultaneously across different networking layers. I introduce two systems, DChannel and DChannelX, which dynamically steer traffic across low-latency and high-bandwidth paths. DChannel operates at the network layer, transparently improving application performance without requiring application-level information or modifications, while DChannelX leverages transport- and application-level information to further optimize traffic steering. To complement these protocol design contributions, I develop CellReplay, an accurate cellular network record-and-replay system that enables reproducible evaluation of networking protocols under dynamic real-world conditions.

PUBLICATIONS

AdvNet: Revealing Performance Issues in Network Protocols by Generating Adversarial Environments
Shehab Sarar Ahmed, William Sentosa, Yinjie Zhang, Yoav Lebendiker, Michael Shnaiderman, Tomer Gilad, Nathan H. Jay, Brighten Godfrey, and Michael Schapira
The 22nd International Conference on emerging Networking EXperiments and Technologies (ACM CoNEXT), 2026

- Infrastructure protocols like Congestion Control (CC) aim to provide reliable performance across a diverse Internet environments such that it is critical to identify and understand scenarios where they perform poorly. AdvNet is a framework that uses Machine Learning (ML) optimization to automatically create adversarial network conditions (e.g., network trace and configurations) to degrade the performance of a target network protocol.

CellReplay: Towards accurate record-and-replay for cellular networks
William Sentosa, Balakrishnan Chandrasekaran, P. Brighten Godfrey, and Haitham Hassanieh
22nd USENIX Symposium on Networked Systems Design and Implementation (**USENIX NSDI**), 2025

- The inherent variability of cellular networks makes it hard to evaluate, reproduce, and debug the performance of networked applications running on these networks. We develop CellReplay, an accurate trace-based emulator

tool that captures real-world variability in cellular networks. Through measurement, we show that cellular networks exhibit latency dynamics that depend on workloads. CellReplay captures these dynamics using a novel interpolation and provides significantly better accuracy ($\approx 2.5\times$) for networked apps (e.g., web browsing and video streaming) compared to the widely used state-of-the-art.

- This paper won NSDI'25 Community Award (best paper whose code and/or data set is made publicly available)

Offloading Head Tracking for Low Power XR: An End-to-End System and Evaluation
Qinjun Jiang, Yihan Pang, William Sentosa, Steven Gao, Muhammad Huzaifa, Jeffrey Zhang, Javier Perez-Ramirez, Dibakar Das, Dave Cavalcanti, P. Brighten Godfrey, and Sarita Adve.
16th ACM Multimedia Systems Conference (**ACM MMsys**), 2025

- We develop RemoteVIO, the first end-to-end extended reality (XR) system that offloads head tracking to a remote server. Our studies with 30 users show that RemoteVIO provides a satisfactory user experience under typical network conditions. Energy tests on real hardware show that RemoteVIO reduces end-to-end system power consumption by up to 20%.

Xrgo: Design and evaluation of rendering offload for low-power extended reality devices
Steven Gao, Jeffrey Liu, Qinjun Jiang, Finn Sinclair, William Sentosa, P. Brighten Godfrey, and Sarita Adve.
16th ACM Multimedia Systems Conference (**ACM MMsys**), 2025

- In a cloud-based VR, reprojection is needed to compensate for an additional latency coming from the network. This paper presents OpenWarp, a mesh-based reprojection technique that generates new frames by warping older frames according to six degrees-of-freedom (DoF) motion. Through user studies, we show that 45% and 75% of users find OpenWarp's quality of experience indistinguishable from on-device rendering under 5G and WiFi networks, respectively.

Is WTSN the missing piece for low latency in general-purpose Wi-Fi?
Milind Kumar Vaddiraju, William Sentosa, Qinjun Jiang, Sarita Adve, Dave Cavalcanti, Dibakar Das, P. Brighten Godfrey, Javier Perez-Ramirez, and Deepak Vasisht.
Twenty-second International Workshop on Mobile Computing Systems and Applications (**ACM HotMobile**), 2025

- Wireless Time-Sensitive Networking (WTSN) offers low-latency, reliable, and time-synchronized data transmission over WiFi. However, it traditionally relies on precise prior knowledge of packet arrival times, which is impractical for many applications. This paper demonstrates that WTSN can achieve low latency without such knowledge by leveraging a multipath design.

Boosting Application Performance using Heterogeneous Virtual Channels: Challenges and Opportunities
Talal Touseef, William Sentosa, Milind Kumar Vaddiraju, Debopam Bhattacherjee, Balakrishnan Chandrasekaran, P. Brighten Godfrey, and Shubham Tiwari.
Twenty-second ACM Workshop on Hot Topics in Networks (**ACM HotNets**), 2023

- This position paper discusses the emergence of the Heterogeneous Virtual Channels (HVCs) across various context (5G, WiFi, WAN), and how to use HVCs in different networking layers (network, transport, and application). HVCs comprise multiple channels or paths, each optimized for specific performance aspects such as throughput, latency, or reliability. We also implement a cross-layer scheme that leverages HVCs to reduce tail latency in real-time video streaming.

DChannel: Accelerating Mobile Applications With Parallel High-bandwidth and Low-latency Channels
William Sentosa, Balakrishnan Chandrasekaran, P. Brighten Godfrey, Haitham Hassanieh, and Bruce Maggs
20th USENIX Symposium on Networked Systems Design and Implementation (**USENIX NSDI**), 2023

- 5G (and beyond) provide differentiated services: high throughput with increased latency (eMBB) and low latency with limited throughput (URLLC). DChannel provides a network-layer solution that dynamically steer packets across both channels, leveraging their combined high bandwidth and low latency benefits. Our experiments show that while URLLC provides only 1% of eMBB's bandwidth, utilizing both channels improves web page load times and mobile app responsiveness by 16–40% compared to using eMBB alone.

cISP: A Speed-of-Light Internet Service Provider
Debopam Bhattacherjee, Waqar Aqeel, Sangeetha Abdu Jyothi, Ilker Nadi Bozkurt, William Sentosa, Muhammad Tirmazi, Anthony Aguirre, Balakrishnan Chandrasekaran, P. Brighten Godfrey, Gregory Laughlin, Bruce Maggs, and Ankit Singla.
19th USENIX Symposium on Networked Systems Design and Implementation (**USENIX NSDI**), 2022

- We explore the design of wide-area networks (cISP) that move data at near light speed in a vacuum. Our cISP design aims to augment the Internet's fiber with free-space microwave wireless connectivity over paths very close to great-circle paths. To achieve ultra low latency, we address various practical challenges ranging from transmission tower availability to packet queuing. Finally, we explore the application-level benefits and present estimates demonstrating that cISP's utility far outweighs its cost.

Accelerating Mobile Applications With Parallel High-bandwidth and Low-latency Channels
William Sentosa, Balakrishnan Chandrasekaran, P. Brighten Godfrey, Haitham Hassanieh, Bruce Maggs, Ankit Singla
The Twenty-second International Workshop on Mobile Computing Systems and Applications (**ACM HotMobile**), 2021

- This position paper is an early version of our work on DChannel (NSDI'23).

Understanding and Auto-Adjusting Performance-Sensitive Configurations
Shu Wang, Chi Li, Henry Hoffman, Shan Lu, William Sentosa, Achmad Imam Kistijantoro
ACM *International Conference on Architectural Support for Programming Languages and Operating Systems* (**ACM ASPLOS**), 2018

- We present SmartConf, a systematic and general control-theoretic framework that automatically tunes and dynamically adjusts performance-sensitive configurations to meet operational constraints while optimizing other performance metrics. We then apply SmartConf to six configuration related performance issues in various software systems (i.e., Cassandra, HBase, HDFS, and MapReduce), and show that it provides better performance than even the best static configuration developers can choose.

WORK EXPERIENCE

Networking and real-time streaming engineer at Apple, USA Sept 2026 - now

- Design and develop the real-time networking stack that powers all Apple's real-time media applications, such FaceTime and Airplay.
- Leverage my research expertise in networking transport and cellular networks to enhance Apple's networking stack.

Research assistant for Project Agora at University of Illinois Urbana-Champaign, USA Aug 2025 - May 2026
https://nsf-breaking-low-agora.illinois.edu

- Participated in an NSF-funded industry-collaborative project (with partners such as T-Mobile, Qualcomm, Cisco, MediView XR, and Peraton Labs, among others) on advancing low-latency wireless networks through vertically integrated R&D.
- Designed and implemented a latency-aware multipath network protocol to support low latency wireless for remote XR medical collaboration.

Graduate research assistant at University of Illinois Urbana-Champaign, USA Jan 2020 - now

- Designed and built networking systems (see the Publications section for more details) in C/C++ on Linux.
- Built infrastructure for systems experiments and automated performance evaluation using Python.
- Led research and engineering projects on low-latency transport protocols and cellular network emulation.
- Mentored 5 graduate and undergraduate students on systems research projects.

Teaching assistant at University of Illinois Urbana-Champaign, USA Jan 2019 - Dec 2019
CS241 (now CS341) - System Programming

- Led lab sessions covering processes, synchronization, scheduling, networking, and memory management in C.
- Awarded UIUC teacher ranks as excellent by their students in Fall 2019.

Data engineer at Bukalapak, Indonesia February 2018 - May 2018

- Optimized Apache HBase and Presto Query engine to support large-scale e-commerce data analytics.
- Tuned distributed storage and query engine configurations to improve throughput and reduce latency.

INVITED TALKS

Accelerating Mobile Applications With Parallel 5G High-bandwidth and Low-latency Channels

- Microsoft Research, Sept 2022, Inviter: Sharad Agarwal
- Embrace.io, May 2022, Inviter: Anuj Jaiswal

TECHNICAL SKILLS

Programming	C/C++, Python, Java
Networking & Transport	TCP/IP, QUIC, HTTP, Linux TC, multipath transport, network measurement
Networked Applications	Web browsing, Video streaming, VR/AR
Systems	Linux networking stack, TUN/TAP interfaces